

WASHINGTON RONALD REAGAN, VA, USA

WMO: 724050

Lat: 38.847N

Lon: 77.035W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	20.7	-3.2	4.6	19.9	1.6	6.0	24.4	26.6	33.9	24.7	32.6	11.9	330	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	94.5	75.6	92.0	74.7	89.6	73.7	78.4	89.1	77.4	87.3	76.3	85.3	10.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	134.1	83.3	74.5	129.0	82.2	73.4	124.2	81.3	41.8	89.6	40.6	87.4	39.6	85.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.0	18.0	12.2	98.1	5.4	3.2	8.3	100.5	5.2	102.3	2.1	104.2	-1.8	106.5
			9.4	81.0	4.9	1.7	5.9	82.3	3.0	83.3	0.3	84.2	-3.3	85.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	36.7	39.5	47.2	57.9	66.8	75.8	80.2	78.9	72.2	60.5	49.5	41.4
	DBStd	17.01	9.83	9.20	9.13	8.18	7.44	5.86	4.77	4.56	6.28	7.76	7.94	8.26
	HDD50	1327	426	312	165	21	0	0	0	0	0	8	105	290
	HDD65	3856	877	714	556	241	71	4	0	0	12	182	466	733
	CDD50	4604	14	18	77	259	520	773	936	895	665	335	90	22
	CDD65	1660	0	0	3	29	126	327	471	430	228	44	2	0
	CDH74	14874	0	2	28	247	946	2893	4902	4015	1588	247	6	0
	CDH80	5635	0	0	3	59	283	1078	2116	1556	489	51	0	0
Wind	WSAvg	8.7	9.4	9.4	10.3	9.7	8.6	8.3	7.8	7.5	8.0	8.1	8.5	8.5
Precipitation	PrecAvg	40.0	2.8	2.6	3.3	2.9	3.8	3.9	4.0	3.5	3.7	3.2	3.0	3.2
	PrecMax	66.3	7.1	5.7	8.5	6.9	10.7	14.0	9.7	9.2	12.4	9.4	7.6	6.5
	PrecMin	28.4	0.4	0.4	0.0	0.0	0.7	0.9	0.7	0.4	0.1	0.0	0.3	0.4
	PrecStd	7.6	1.3	1.4	1.5	1.3	2.0	2.8	2.1	2.0	2.8	2.1	1.8	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	70.8	77.9	85.9	90.2	95.9	98.8	96.6	93.4	85.7	74.0	68.5
		MCWB	57.7	58.5	61.1	66.6	72.2	75.7	76.5	76.5	73.4	71.5	62.2	60.4
	2%	DB	60.6	63.1	71.2	80.9	86.4	91.8	95.2	93.3	89.2	80.1	68.7	62.3
		MCWB	55.2	53.5	57.6	63.9	70.8	74.2	75.9	75.5	72.2	69.1	59.8	56.8
	5%	DB	55.6	57.2	66.5	76.0	83.1	89.0	92.4	90.5	85.4	76.2	65.2	57.5
		MCWB	50.1	48.0	55.0	62.0	69.2	73.3	75.4	74.1	70.9	66.6	57.9	52.5
	10%	DB	50.2	52.6	61.5	71.7	79.5	86.2	89.8	87.9	82.2	72.5	61.8	53.1
		MCWB	44.4	45.8	51.7	59.6	67.4	72.1	74.5	73.1	70.1	64.4	55.0	47.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	61.8	63.7	70.3	75.1	78.8	80.6	80.0	77.6	74.5	66.1	63.1
		MCDB	63.8	68.6	74.0	79.6	86.2	90.2	91.7	91.3	86.7	79.8	69.9	66.7
	2%	WB	56.5	55.3	60.2	67.1	73.0	77.1	78.8	78.1	75.8	71.5	62.6	58.0
		MCDB	60.7	61.5	68.4	75.9	83.2	87.0	89.7	88.7	83.5	77.6	66.8	61.5
	5%	WB	50.8	50.3	56.9	64.3	71.1	75.9	77.6	76.9	74.4	69.0	59.4	53.4
		MCDB	54.9	55.1	64.8	73.4	80.4	85.2	87.9	86.2	81.3	74.3	63.9	57.0
	10%	WB	45.6	46.2	53.0	61.7	69.2	74.6	76.5	75.5	72.8	65.8	56.1	48.4
		MCDB	49.6	52.1	60.0	70.4	77.1	83.0	86.0	83.2	79.3	71.0	60.4	52.3
Mean Daily Temperature Range	5% DB	MDBR	13.5	15.0	16.9	18.3	17.0	16.2	15.7	15.5	15.4	15.7	15.1	13.0
		MCDBR	20.6	22.7	24.0	24.3	20.9	19.2	18.7	18.4	19.0	19.3	19.7	18.8
		MCWBR	16.1	15.3	14.3	12.0	9.8	7.7	6.8	6.8	7.7	10.1	12.9	15.1
	5% WB	MCDBR	19.4	19.9	22.2	21.2	18.9	17.1	16.4	16.1	15.1	16.2	16.8	17.3
		MCWBR	16.9	15.5	15.4	12.3	9.4	7.7	6.9	6.9	7.6	10.4	13.4	15.7
Clear-Sky Solar Irradiance	taub	0.309	0.323	0.358	0.404	0.445	0.502	0.524	0.518	0.430	0.363	0.325	0.311	
	taud	2.488	2.430	2.396	2.285	2.202	2.083	2.041	2.039	2.261	2.453	2.516	2.519	
	Ebn at Noon	276	286	285	277	267	251	244	241	258	268	267	267	
	Edh at Noon	25	30	35	41	45	51	53	52	39	29	24	23	
All-Sky Solar Radiation	RadAvg	663	930	1236	1563	1763	1947	1908	1702	1387	1040	755	582	
	RadStd	55	103	90	118	197	148	98	90	161	115	64	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	-178	+301	+193
(47)	Regional (74 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+133

Nomenclature: See separate page